

Water

INTRODUCTION

Of all of the nutrients consumed by dairy cattle, water is consumed in the greatest amounts. Water is essential for life and only follows oxygen in importance. In addition, no nutrient is found in greater concentrations in either the body of a mature cow (~65 percent), her fetus (~80 percent), or her milk (85 percent), and the body recycles no nutritional element more so than water. Even small changes in body water can result in important changes in animal health and performance. Water flux in a lactating dairy cows averages about 30 percent (Beede, 2012), which is greater than any other domesticated ruminant (Woodford et al., 1984a).

Water possesses unique physical and chemical properties that allow it to act as a solvent and to support life. Two important characteristics of water are that (1) the molecule is electrically polar and that (2) an unshared pair of electrons on the oxygen atom can bond with a hydrogen (H) atom on another molecule, creating a hydrogen bond. A consequence of these two characteristics is that water molecules are attracted to each other. Water also possesses several properties that contribute to the effectiveness in regulating body temperature (Denny, 1993). First, compared to virtually all other liquids at room temperature, water has a high specific heat (at 0°C, it is 4,218 J kg⁻¹ K⁻¹). Thus, to alter its temperature, large amounts of heat need to be added or removed. In addition, the thermal conductivity of water is 0.565 W m⁻¹ K⁻¹ at 0°C and aids the dissipation of heat from the body (Denny, 1993). Water has a high latent heat of vaporization, thus allowing for the evaporation of water from the skin and respiratory tract. This characteristic of water creates a notable route of heat loss for cattle (Monteith, 1972; Squires, 1988). The true number of functions served by water is not fully known, but major functions include regulating body temperature, supporting intermediary metabolism by acting as a solvent to dissolve substances, transporting nutrients and metabolites throughout the body, and eliminating waste materials in urine, feces, and respiration. Last, water serves

as a lubricant in joints and in many organs. In cerebrospinal fluid, water acts as a cushion for the brain and brain and spinal tissue (Roubicek, 1969). Given the number of functions related to water, restricting water intake results in rapid but often reversible reductions in feed intake and milk yield (Burgos et al., 2001).

POOLS OF BODY WATER

Total Body Water

Total body water (TBW) is composed of intracellular fluid water and extracellular fluid water (ECW). ECW can be broken down into blood plasma water, interstitial water, and transcellular water (Hix et al., 1959; Murphy, 1992). The intracellular pool is the largest pool at approximately 40 percent of body weight (BW; Murphy, 1992). The ECW pool includes water contained in saliva, plasma, and interstitial fluid. The plasma volume of water in lactating cows is about 6.4 percent of BW (Woodford et al., 1984a). Because milk contains roughly 85 percent water, the ECW pool, which includes milk, is proportionally large. The proportion of ECW within the gastrointestinal tract is located mostly in the rumen and is about 65, 62, and 61 percent of total ECW in cows on 7, 63, and 269 days postpartum, respectively (Andrew et al., 1995).

Water enters the reticulo-rumen pool through saliva, through swallowed water, and by consuming feed containing water (Appuhamy et al., 2014). A portion of drinking water may pass directly into the abomasum through the esophageal groove (Woodford et al., 1984b) but likely is only 11 to 22 percent of the total drinking water intake (Woodford et al., 1984b; Cafe and Poppi, 1994).

Empty Body Water

Empty body water (EBWtr) is the proportion of water contained in the animal minus that contained in the ingesta.

In general, EBWtr decreases with increasing body fat content (Maeno et al., 2013); thus, in calves, EBWtr is approximately 70 percent (Chapman et al., 2017), and as they age, EBWtr decreases before reaching a relatively constant value on physiological maturity (Lohman, 1971). TBW is lower for fat dry cows and higher in lactating cows (Aschbacher et al., 1965; Murphy, 1992). Andrew et al. (1995) reported that the EBWtr content at 7, 63, and 269 days postpartum is 59, 66, and 60 percent of BW, which are similar to more recent data for lactating cows (64.7 ± 3.02 percent of BW; Agnew et al., 2005).

WATER BALANCE

Cattle lack the ability to store bulk volumes of water for extended periods of time (Dukes, 1955). Provided that water is available throughout the day, the volume of water in the body remains relatively consistent (Reece, 2004). When needed, the animal gains water by ingestion via drinking and consuming feed containing water and through metabolic oxidation. Water is lost via feces, urine, and sweating and respiration, while a small volume is also lost through saliva (Holter and Urban, 1992). When lactating, milk is a major route of water loss; however, the proportion of water in milk is highly regulated and does not fluctuate with changes in whole-animal fluid balance (Olsson, 2005).

Water Intake

Thirst and Drinking Behavior

Physiologically, homeostatic factors regulate pH, osmotic pressure, and acid-base balance and are modulated by the movement of ions such as sodium (Na), potassium (K), chloride (Cl⁻), and bicarbonate through intracellular and extracellular fluids containing water and solutes (Reece, 2004; Hogan et al., 2007). Consequently, the gain and loss of body water plays a major role in maintaining homeostasis.

Thirst is defined as a “longing or compelling desire to drink” and is stimulated by either extracellular or cellular dehydration (Hogan et al., 2007). Thirst may be triggered by a reduction in salivary secretion and dryness of the throat and mouth. The hypothalamic region of the brain controls thirst and drinking behavior and is mediated by angiotensin II (Hogan et al., 2007). Osborne et al. (2002) observed that cows consumed 40 percent of their daily water intake within 2 hours of each feeding and milking time. Cattle generally use the musculature found in their cheeks to create suction that draws water upward. This water is then directed by the tongue and transported intraorally (Reis et al., 2010). Drinking is composed of one cycle in which one aliquot of water is first sucked up and then swallowed (Hiiemae and Crompton, 1985). The frequency of water consumption varies depending on an array of factors that most notably include water availability,

and lactating cows should have access to unlimited amounts of water throughout the day (Radostits and Blood, 1985), especially under hot conditions (West, 2003). The frequency of discrete water consumption episodes varies from five for cows in late lactation (Jago et al., 2005) to almost eight in low stocking environments (Cardot et al., 2008). The frequency of water consumption is reduced when animals are on pasture compared to those in confinement (Jago et al., 2005). In some cases (e.g., winter), beef cattle have been known to survive as many as 3 to 5 days without water (Siebert and Macfarlane, 1975; Squires, 1988). In lactating dairy cattle, the total time spent drinking ranges between 13 and 17 min/d (Thomas et al., 2007). The volume of water consumed is positively correlated with the animal’s standing of social dominance within the herd (Andersson and Lindgren, 1987). Pinheiro Machado Filho et al. (2004) observed that cows consume more water from large deeper troughs.

Free Water Intake

Free water intake (FWI) is defined as water that is consumed directly from a water store or watering device.

Lactating Cows

Several factors that affect daily FWI by dairy cows have been identified. Table 9-1 is a list of published equations used to predict FWI (kg/d) in lactating and dry dairy cows. The previous report (NRC, 2001) recommended the equation of Murphy et al. (1983); however, newer equations that attempt to identify more factors and account for more variation have been developed. Controlled studies to quantify the effects of ambient temperature, temperature humidity index (THI), solar radiation, relative humidity, wind speed, and precipitation on FWI are needed, but some of these variables are used in predictive equations (see Table 9-1). In general, because mean and minimum and maximum daily temperatures are closely correlated, one measure is probably suitable for predictive purposes (Murphy et al., 1983). Recently, Appuhamy et al. (2016) evaluated published equations used to predict FWI and developed and evaluated new predictive equations (see Table 9-1). Recommendations of Appuhamy et al. (2016) and those adopted for this report are that when reliable estimates of dry matter intake (DMI) are available, Equation 9-1 (see Table 9-1) be used to predict FWI while Equation 9-2 (see Table 9-1) be used when reliable estimates of DMI are not available. Both Equations 9-1 and 9-2 are unique to others because they include dietary K. K has been shown to positively affect both water consumption (Meyer et al., 2004; Fraley et al., 2015) and ruminal liquid passage rates (Fraley et al., 2015). Given the lack of data available to develop these equations, seven alternative equations are also listed. Appuhamy et al. (2006) noted that the equation of Murphy et al. (1983) and Meyer et al. (2004) both required DMI and, when evaluated, performed well. In

TABLE 9-1 Equations Used to Predict FWI (kg/d) in Dairy Cattle^a

Equation	Reference	Models for Predicting FWI (kg/d) ^b
Recommended equations Lactating cows Equation 9-1 Equation 9-2	Appuhamy et al. (2016) Appuhamy et al. (2016)	$= -91.1 + (2.93 \times \text{DMI}) + (0.61 \times \text{DM}\%) + (0.062 \times \text{NaK}) + (2.49 \times \text{CP}\%) + (0.76 \times \text{TMP})$ $= -60.2 + (1.43 \times \text{Milk}) + (0.064 \times \text{NaK}) + (0.83 \times \text{DM}\%) + (0.54 \times \text{TMP}) + (0.08 \times \text{DIM})$
Alternative equations	Murphy et al. (1983) Meyer et al. (2004) Murphy et al. (1983) Holter and Urban (1992) Khelil-Arfa et al. (2012) Appuhamy et al. (2014) Little and Shaw (1978) Stockdale and King (1983) Castle and Thomas (1975) Khelil-Arfa et al. (2012) Dahlbom et al. (1998)	$= 16.0 + (1.58 \times \text{DMI}) + (0.90 \times \text{Milk}) + (0.05 \times \text{NaI}) + (1.20 \times \text{mnTMP})$ $= -26.1 + (1.30 \times \text{Milk}) + (0.406 \times \text{NaI}) + (1.516 \times \text{TMP}) + (0.058 \times \text{BW})$ $= -23.0 + (2.38 \times \text{DMI}) + (0.64 \times \text{Milk})$ $= -32.4 + (2.47 \times \text{DMI}) + (0.60 \times \text{Milk}) + (0.62 \times \text{DM}\%) + (0.091 \times \text{JD}) - (0.00026 \times \text{JD}^2)$ $= -77.6 + (3.22 \times \text{DMI}) + (0.92 \times \text{Milk}) - (0.28 \times \text{CONC}\%) + (0.83 \times \text{DM}\%) + (0.037 \times \text{BW})$ $= -34.6 + (2.75 \times \text{DMI}) + (0.84 \times \text{Milk}) + (2.32 \times \text{Ash}\%) + (0.27 \times \text{DM}\%)$ $= 12.3 + (2.15 \times \text{DMI}) + (0.73 \times \text{Milk})$ $= -9.37 + (2.30 \times \text{DMI}) + (0.53 \times \text{DM}\%)$ $= -15.3 + (2.53 \times \text{Milk}) + (0.45 \times \text{DM}\%)$ $= -41.1 + (1.54 \times \text{Milk}) - (0.29 \times \text{CONC}\%) + (0.97 \times \text{DM}\%) + (0.039 \times \text{BW})$ $= 14.3 + (1.28 \times \text{Milk}) + (0.32 \times \text{DM}\%)$
Recommended equation Dry Cows Equation 9-3	Appuhamy et al. (2016)	$= (1.16 \times \text{DMI}) + (0.23 \times \text{DM}\%) + (0.44 \times \text{TMP}) + (0.061 \times \text{TMPC}^2)$
Alternative equations	Appuhamy et al. (2016) Holter and Urban (1992)	$= (0.69 \times \text{DMI}) + (0.28 \times \text{DM}\%) + (0.85 \times \text{TMP})$ $= 10.34 + (0.230 \times \text{DM}\%) + (2.21 \times \text{DMI}) + (0.0394 \times (\text{CP}\%)^2)$

^aAdapted from Appuhamy et al. (2016).

^bDMI (kg/d), BW (kg), Milk = milk yield (kg/d), DM% = dry matter percentage of the diet, CONC% = concentrate content of the diet (% of DM), CP% = dietary CP content (% of DM), Ash% = dietary total ash content (% DM), NaI = sodium intake (g/d), TMP = daily average ambient temperature (°C), mnTMP = daily minimum ambient temperature (°C), JD = Julian day, TMP = daily mean ambient temperature (°C), TMPC² = $(\text{TMP} - 16.4)^2$, and NaK = sum concentration of Na and K in the diet, milliequivalent/DM kg, $(\% \text{Na}/0.023) + (\% \text{K}/0.039) \times 10$.

general, measures to increase consumption of water should be encouraged, but water intake alone should not be used to evaluate the effects of water quality. One cannot assume underconsumption is a result of poor water quality as it may be reduced in response to other factors such as poor health or production as well as access to watering devices (Kononoff et al., 2017).

Dry Cows

Fewer equations exist to predict FWI in dry cows (see Table 9-1), and further research in this area is needed to test current predictive equations. Equations for dry cows are based mostly on DMI, dry matter (DM) concentration of the diet, and ambient temperature; however, factors such as K likely affect water intake, but data are lacking. For dry cows, Equation 9-3 (see Table 9-1) is recommended (Appuhamy et al., 2016), but published studies measuring FWI in dry cows are lacking, and this prediction will likely be improved with more data. Because an independent data set was not available, Appuhamy et al. (2016) did not compare Equation 9-3 to the alternative equation of Holter and Urban (1992) also listed in Table 9-1. Equation 9-3 is recommended because data used to develop it represented a greater range of DMI as well as environmental and diet conditions, but it is possible that the

alternative equation of Holter and Urban (1992) could predict FWI as well as or better than Equation 9-3.

Calves and Heifers

Water should be provided free choice to calves, including those being fed a liquid diet (Drackley, 2008). Kertz et al. (1984) observed that weight gain was reduced by 38 percent and starter intake was reduced by 31 percent when calves had restricted access to water. Currently, models to predict FWI in young calves are not available, and published data are scarce, thereby precluding development of a model. Some studies have reported low FWI during the period before weaning (de Passille et al., 2011), but most observe that in early life, FWI is approximately 0.75 to 1 kg/d (Thomas et al., 2007; Wickramasinghe et al., 2019) and increases with age (Wenge et al., 2014). By 20 days of age, FWI increases dramatically (Kertz et al., 1984), and this increase in FWI occurs in parallel with reductions in feeding of milk replacer and increasing starter intake. DMI (in the form of calf starter) is likely directly related to FWI in young calves, and calves may require four times greater FWI than DMI or an FWI to DMI ratio of 4:1 (kg basis) (Kertz, 2014). Quigley et al. (2006) reported that prior to weaning, FWI/DMI was 2:1 but that this increased to 4:1 after full weaning. The amount

of water consumed prior to weaning is also a function of liquid feed consumption, and when liquid feeding rates are high, FWI/DMI may be less than 2:1 (Wickramasinghe et al., 2019). In addition, these investigators determined that withholding water until 17 days of age reduced milk intake, as well as BW and heart girth at 5 months of age compared to calves given access to water at birth. Increases in FWI in calves are also associated with increased environmental temperatures, feed restriction, increased water temperature in cold environments (Huuskonen et al., 2011), increased starter intake (Kertz, 1984; Wenge et al., 2014), and development of ruminal fermentation (Abe et al., 1999a,b). Providing fresh warm water is important in calves suffering from diarrhea (McGuirk, 2008) because calves increase FWI by 25 to 50 percent when suffering from diarrhea (Jenny et al., 1978). Data on FWI in older growing heifers are limited, and no equations have been developed to predict their FWI. Equations outlined by NASEM (2016) to predict FWI in growing feedlot cattle do not appear to be accurate based on limited data such as that published by Zanton and Heinrichs (2016).

Ambient Temperatures and Free Water Intake

The increase in FWI with increasing temperatures is well known (NRC, 1981), but the response is variable across individual animals and locations (Arias and Mader, 2011). During hot weather, the increase in water consumption is believed to be a response to the need to support evaporative and respiratory heat losses (Pereira et al., 2014). If not properly restored, water located within the vascular and extracellular compartments may be disrupted, leading to interference with osmotic pressure and blood pressure. Such physiological changes can ultimately threaten thermoregulation and cardiovascular function (Silanikove, 1994). McDowell et al. (1969) observed that FWI is 29 percent greater when a Holstein cow is housed at 32°C compared to being housed in temperatures between 15°C and 24°C. In a temperature-controlled study (temperature mean = 8.6±7.1; minimum = -5.6°C; maximum = 23.3°C), for each degree Celsius increase in ambient temperature, FWI increased by 1.5 kg (Meyer et al., 2004). These investigators concluded that daily mean and minimum or maximum environmental temperatures are highly correlated with FWI, and each may influence FWI. In another controlled study, increasing ambient temperature from 15°C to 28°C increased FWI by 10 and 42 percent for lactating and dry cows, respectively (Khelil-Arfa et al., 2014). Evaporative losses were estimated as a proportion of DMI, and those losses were compensated for by an increase in FWI.

Other Factors Affecting Free Water Intake

In general, and even in warm environmental temperatures, cattle likely prefer warm over cool water (30°C versus <14°C) and cattle prefer water between 20°C and 28°C

(Lanham et al., 1986). In hot, arid climates, a preference for cool water exists (Challis et al., 1987). In a study conducted in Canada and over four seasons, water was offered at either ambient temperature (7°C to 10°C) or warmed (30°C to 33°C), and FWI increased with water temperature. The greatest change in intake response was observed in the winter (5.9 percent) while the lowest change in response (2.8 percent) was observed in the spring (Osborne et al., 2002). Cows in this study were not under heat stress. In a similar study using bull calves, water was offered at either cool (6°C to 8°C) or warmed (16°C to 18°C) temperature, and FWI increased with water temperature during both the preweaning and postweaning stages (Huuskonen et al., 2011). In the summer months, chilling water has increased FWI of lactating dairy cattle and reduced respiration rates and body temperature (Lanham et al., 1986; Milam et al., 1986). In grazing cattle, warm environmental conditions play a major role on both drinking behaviors and FWI. The number of drinking bouts increases with THI, but at very high THI, the number of bouts decreases, possibly indicating an inability to thermoregulate in these conditions (Pereyra et al., 2010). Estrus can decrease FWI in lactating cows (Reith et al., 2014); however, flavoring agents (orange or vanilla) did not affect fluid water intake (Thomas et al., 2007).

WATER LOSSES

Milk Losses

Holter and Urban (1992) summarized four energy balance trials with 329 lactating Holstein cows housed at 18°C and observed that water losses through milk averaged 34 percent and ranged from 19 to 52 percent of total water intake (TWI; fluid plus feed water). Cows housed in a climatic chamber at 15°C had water losses through milk that averaged 24 percent of TWI, but this was reduced to 21 percent when the animals were housed in high temperatures (Khelil-Arfa et al., 2014; see Figure 9-1).

Fecal and Urinary Losses

For cows producing 23 kg of milk, fecal water contributed 61 percent of the total manure water (Appuhamy et al., 2014). The proportion of water lost in the feces when expressed as the percentage of TWI can be as low as 30 percent in lactating cows (McDowell et al., 1969) and as high as 44 percent (Khelil-Arfa et al., 2014) in thermoneutral conditions but decreased to 35 percent in warmer temperatures (see Figure 9-1).

The pituitary hormone, antidiuretic hormone (ADH), also known as vasopressin, largely regulates the excretion of water by the kidney. The release of ADH is likely governed by plasma osmoconcentration, but it may also be stimulated by pain, exercise, or psychological stress. When the animal is deprived of water, the concentration of ADH in

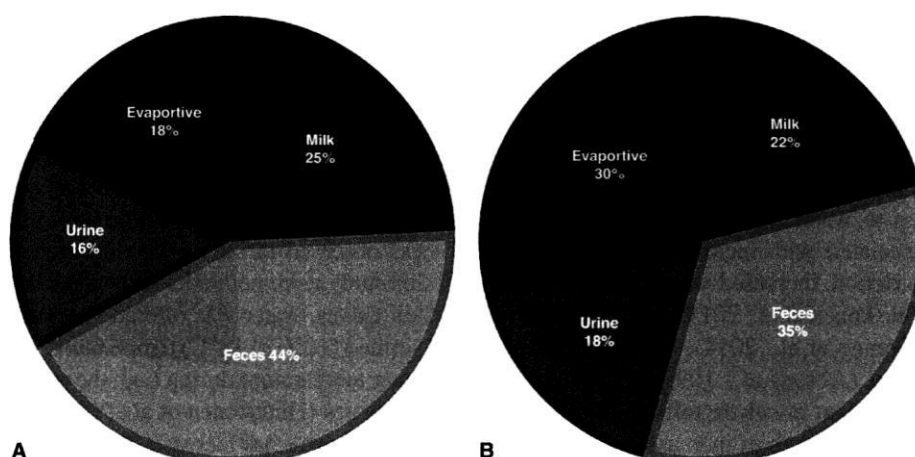


FIGURE 9-1 Water losses as reported by percentage of TWI (water acquired through drinking and feed consumption) by lactating dairy cows housed in (A) thermoneutral (15°C, TWI = 108 kg) and (B) high temperatures (28°C, TWI = 113 kg) as reported by Khelil-Arfa et al. (2014). NOTE: Due to imbalance in metabolic water, retained water, and analytical error, values do not sum to 100 percent.

the blood increases, resulting in a reduction in urine volume. Conversely, when the animal is in excessive positive fluid balance, the concentration of ADH is reduced in the blood, and water excreted in the urine is increased until it is similar in concentration to that of plasma (Reece, 2004). Urinary losses of water have been reported to range between 11 and 21 percent of TWI (McDowell et al., 1969; Holter and Urban, 1992; Dahlbom et al., 1998). In a study evaluating the effect of increasing ambient temperature (15°C to 28°C) and so- dium bicarbonate (0.20 percent DM and 0.50 percent DM), Khelil-Arfa et al. (2014) observed urine losses increased from 15 to 21 percent in lactating cows as temperature and sodium bicarbonate increased. These effects did not occur with dry cows.

Evaporative Loss

Water lost through evaporation increased from 18 to 30 percent of TWI when lactating cows moved from thermo- neutral to higher-temperature conditions (Khelil-Arfa et al., 2014; see Figure 9-1). In dry cows, the response was 28 to 44 percent. Differences between lactating and dry cows may be due to a change in fractionation of the body water pool (Abeni et al., 2015). The efficiency of evaporative losses from the skin is also affected by the thickness, length, and color of the haircoat (Gebremedhin et al., 2008).

Sweat Losses

Sweating is an active process, which is triggered by an increase in body core temperatures and involves the secre- tion of fluid by the sweat glands (NRC, 2007). During this process, heat along with water is lost from the surface of the

skin (Gebremedhin and Wu, 2002). To dissipate heat, dairy cattle sweat in two different ways (Bemabucci et al., 2010). The first is insensible sweating, in which, unless relative humidity is 100 percent, sweat leaves the body constantly. The second is thermal sweating and serves as the principal mechanism of cooling with increasing temperatures. The vaporization of 1 L of water or sweat requires 0.58 Meal (2.42 MJ) (Bemabucci et al., 2010). Jersey cows have a sweating rate of 189 ± 84.6 g/m²-h, while mostly black or mostly white Holsteins have a sweating rate of 414 ± 158.7 g/m²-h or 281 ± 119.4 g/m²-h, respectively (Gebremedhin et al., 2008). These observations support the suggestion that Jersey cattle are more heat tolerant than Hol- stein cattle and that mostly black Holstein cattle possess higher solar absorption characteristics than mostly white Holstein cattle.

Respiratory Losses

Cattle also lose water through respiration. This type of loss is enhanced and facilitated through polypnea or the behavior known as panting (Gaughan and Mader, 2014). Research conducted in Missouri (Kibler and Brody, 1949, 1950, 1952, 1954) attempted to quantify heat loss through different routes. In summarizing these observations, Brouk et al. (2003) noted that at temperatures above 21°C, heat was primarily lost through moisture evaporation from the skin and lungs. However, in animals that were not cooled and as temperatures exceeded 32°C, over 85 percent of the total heat dissipation occurred through vaporization of water from the body surface and lungs. Respiratory water loss (RWL) at different air temperatures (Ta) and relative humidity (RH) increases with rising Ta but declines with increasing RH with no interaction (Berman, 2006). Using data from climate-

controlled chambers, Berman (2006) developed the following model to predict RWL (g h^{-1}) at different T_a ($^{\circ}\text{C}$) and RH (%):

$$\text{RWL} = 0.41 - 0.02 \times T_a + 0.0005 \times T_a^2 - 0.004 \times \text{RH} + 0.00004 \times \text{RH}^2 \text{ (Equation 9-4)}$$

Water Restriction and

Dehydration

The metabolic use and loss of body water is a continuous process, but consumption is not. Because of this all animals, but especially lactating cows housed in confinement, should have almost continuous access to clean water. Dehydration of as little as 10 percent of TBW may have serious implications on health, while the loss of 15 to 20 percent may be fatal (Beede, 2012). When water intake was restricted to 50 percent expected voluntary intake for 4 days, milk yield dropped by 74 percent, but when cows were allowed to consume 90 percent of expected water intake for 14 days, milk yield only decreased 3 percent (Little et al., 1980). In both situations, water restriction caused significant changes in blood composition, with all analytes increasing in concentration. These findings suggest a reduction in blood volume and hemoconcentration.

WATER QUALITY

Water Can contain dissolved minerals, organic compounds, and microorganisms that may affect milk production and animal health. In 2005, NRC (2005) published guidelines on mineral tolerances from many sources, including water. The committee acknowledged that although there have been substantial advancements of analytical methods to measure minerals in water, information is limited on how many of these minerals affect animals. Studies evaluating the effects of water quality on dairy cattle are limited, and as a result, guidelines are usually extrapolated across species. Notably, guidelines for water quality for humans are often included that are particularly conservative. Because the quality of water may change over time and season, water should be sampled periodically during different seasons and assayed. Maintaining historical data can be useful in identifying subtle changes in water quality. Specific sampling protocols have been developed for water because often it contains only trace amounts of some minerals, and microbiological testing requires aseptic sampling. Commercial testing laboratories can provide accepted sampling protocols, and usually they will provide proper sample containers. To evaluate water for dairy cows, the water needs to be sampled at the point of consumption; however, sampling at different points in the water supply change can help identify sources of contamination (Dege, 2011). Table 9-2 lists thresholds of what can be considered upper potentially concernable concentrations for drinking water in cattle. Information in this table should be used cautiously as there is a general lack of research information around many components (Beede, 2012). Nonetheless,

when water samples contain constituents greater than what is listed in the table, the taste and odor of water may be affected. In addition, diet modifications may need to be made to avoid mineral problems or toxicities described in Chapter 7.

Total Dissolved Solids and Salinity

Total dissolved solids (TDS) are an inexact measure of inorganic constituents dissolved in a water sample because it may also include organic compounds. The term “salinity” is sometimes used to refer to TDS; however, in this usage, salinity refers to all dissolved salts (e.g., magnesium sulfate, sodium bicarbonate, and sodium chloride). TDS are the concentration of total ions present in water but do not identify and quantify individual components of that water sample. Consequently, its value as a quality indicator for water is limited and should be interpreted with caution. The nature of TDS is influenced by the local geology, but the primary ions usually found in water are carbonate, bicarbonate, Cl^- , fluoride, sulfate, phosphate, nitrate, calcium (Ca), magnesium (Mg), Na, and K (NRC, 1974). Cl^- is the ionized form of chlorine (Cl), and those two elemental forms have very different effects both in the rumen and on animal tissues.

Although the effect of varying TDS on milk production has been investigated, effects are likely influenced by the ions used to alter TDS. Therefore, effects of TDS on dairy cattle across studies are variable (Challis et al., 1987; Bahman et al., 1993; Valtorta et al., 2008; Shapasand et al., 2010). Nonetheless, general guidelines have been established (see Table 9-3).

In a study with grazing Holstein cows producing about 24 kg/d of milk, drinking water with 1,000, 5,000, or 10,000 mg/L TDS did not affect milk production or composition (Valtorta et al., 2008). In that study, TDS were increased by adding Na and calcium chloride, Mg and sodium sulfate, and sodium bicarbonate. Bahman et al. (1993) reported no difference in milk yields (averaged about 22 kg) between cows fed water with about 450 or 3,600 mg/L TDS (difference was mostly sulfate, Cl^- , Na, Ca, and Mg). Conversely, “high-producing” cows (based on 1988 standards; actual production not given) had about 25 percent lower lactation persistency when consuming water with 4,100 mg/L TDS compared to cows consuming water with 450 mg/L TDS; however, water intake was actually greater for the high-TDS water (Wegner and Schuh, 1988). Similarly, Challis et al. (1987) reported that reducing TDS from approximately 4,400 mg/L to 441 mg/L increased milk production from about 25 kg/d to 34 kg/d. In that study, TDS were elevated mostly by sulfate, but the high-TDS water also had more Ca, Mg, Na, and Cl^- than the low-TDS water.

Effects of drinking saline water (i.e., water with high TDS from predominantly sodium chloride [NaCl]) are more consistent. When TDS were increased from about 200 to

TABLE 9-2 Drinking Water Standards for Humans and Upper Potentially Concernable Concentrations for Cattle^{a, b}

	U.S. EPA Enforceable or Secondary	Human MCL ^j	This Publication Potentially Concernable Concentrations for Cattle See Table 9-3
TDSs ^c mg/L	Secondary	500	—
pH	Secondary	6.5-8.5	—
Nitrate-N (NO ₃ -N), mg/L	Enforceable	10 ^f	See Table 9-5
Nitrite - N (NCL ₂ N), mg/L	Enforceable	1.0	—
Sulfate (SO ₄ ²⁻), mg/L Secondary	—	250 ^g	Calves = 500 Adult cattle = 1,000
Chemical, mg/L unless otherwise listed			
Aluminum	Secondary	0.05-0.20	5.0-10.0 (Beede, 2012)
Arsenic	Enforceable	0.01	0.20 (NRC, 2005)
Barium	Enforceable	2.0	>10 (Beede, 2012)
Boron	—	—	150 (NRC, 2005)
Cadmium	Enforceable	0.005	0.01-0.05 (NRC, 2005; Beede, 2012)
Calcium	—	—	— ^h
Chloride	Secondary	250	300 (Beede, 2012)
Chlorine (Cl ₂)	Enforceable	4.0 ⁱ	—
Chromium	Enforceable	0.1	0.1-1.0 (NRC, 2005; Beede, 2012)
Cobalt	—	—	1.0 (NRC, 2005)
Copper	Enforceable	1.3	1.3 (EPA, 2009)
Copper	Secondary	1.0	—
Fluoride	Secondary	2.0	2.0 (NRC, 2005)
Iron	Secondary	0.3	0.40 (Beede, 2012)
Lead	Enforceable	0.015	0.05-0.10 (NRC, 2005)
Magnesium	—	—	— ^h
Manganese	Secondary	0.05	0.50 (Beede, 2012)
Mercury	Enforceable	0.002	0.01 (Beede, 2012)
Molybdenum	—	—	0.06 (Beede, 2012)
Nickel	—	—	1.0 (Beede, 2012)
Phosphorus	—	—	—
Potassium	—	—	— ^j
Selenium	Enforceable	0.05	0.05 (Beede, 2012)
Silver	Secondary	0.10	0.05 (Beede, 2012)
Sodium	—	—	300 (Beede, 2012)
Vanadium	—	—	0.10 (NRC, 2005; Beede, 2012)
Zinc	Secondary	5.0	5.0-25.0 (NRC, 2005; Beede, 2012)

NOTES: There is a general lack of research information around many components; caution for use of this table should be exercised, when water samples contain constituents greater than what is listed in the table, the taste and odor of water may be affected and/or diet modifications may need to be made to avoid problems or toxicities.

^aRanges listed reflect a lack of information.

^bProblems may occur when the following are observed (Beede, 2012): fluoride >2.4 mg/L may result in mottling, manganese >0.05 mg/L may affect taste, and sodium >20 mg/L may affect veal calves.

^cEnforceable standard (EPA, 2009). Secondary are nonenforceable guidelines regarding contaminants that may cause cosmetic or aesthetic effects in drinking water for humans. The EPA recommends secondary standards to water systems but does not require systems to comply (EPA, 2009).

^dMCL = Maximum contaminant level for humans, the highest concentration of a contaminant that is allowed in drinking water. MCL is only associated with enforceable standards (EPA, 2009).

^eTDS = total dissolved solids.

^fEquivalent to 44 mg/L of nitrate (NO₃).

^gSulfate sulfur (SO₄²⁻-S) = sulfate (SO₄²⁻) x 0.333.

^hPotentially concernable concentrations for cattle for calcium and magnesium are unknown, but these may affect total dissolved solids (TDS): calcium >500 mg/L and/or magnesium >125 mg/L have been suggested to be concentrations worthy of further evaluation (Beede, 2012).

ⁱMaximum residual disinfectant level for humans, the highest level of a disinfectant allowed in drinking water.

^jPotentially concernable concentrations for cattle for potassium are unknown but concentrations >20 mg/L in drinking water fed to dry cows may warrant further evaluation because of its impact on dietary cation-anion difference.

2,500 mg/L via addition of NaCl, milk yield decreased from 34.8 to 32.9 kg/d (Jaster et al., 1978). Solomon et al. (1995) found that Holstein cows that consumed water with TDS of about 440 mg/L produced more fat-corrected milk (31.6 versus 29.8 kg/d) than cows that consumed water with TDS of about 1,500 mg/L (NaCl was mostly increased, but water

also differed in sulfate, Ca, and Mg). Conversely, Arjoman-dfar et al. (2010) observed no effect on milk yield (averaged 35 kg/d) when TDS were reduced from 1,400 to 570 mg/kg through desalination (the high-TDS water contained pre-dominantly NaCl but also contained higher concentrations of Ca, Mg, and sulfate).

TABLE 9-3 Guidelines for TDS in Water for Dairy Cattle Consumption^a

TDS (mg/L)	Comments
<1,000	Safe and should pose no health problems.
1,000-2,999	Generally safe but may cause a mild, temporary diarrhea in animals not accustomed to the water.
3,000-4,999	Water may be refused when first offered to animals or cause temporary diarrhea. Animal performance may be less than optimum because water intake is not maximized.
5,000-6,999	Avoid these waters as a source of drinking water, may result in reductions in milk production.
>7,000	These waters should not be fed to cattle. Health problems and/or poor production will result.

^aIn general, TDS alone are not adequate to characterize drinking water of cattle, and it is further suggested that specific salt components and bacteriological measures are also needed.

SOURCE: NRC (1974).

A likely reason for the mixed responses to reducing TDS in water is the ionic makeup of the water. For example, high intakes of sulfate and CL are detrimental to milk production during summer months (Sanchez et al., 1994). Furthermore, high concentrations of these minerals in water will likely decrease the dietary cation-anion difference (DCAD)

consumed by the cow, which can reduce intake and milk yield (see Chapter 7). The DCAD is usually calculated as $(Na + K) - (Cl + S)$, where mineral concentrations are expressed as mEq/kg. Including Na or K supplied by water into that equation generally does not alter DCAD because the counterion of the cation is usually CL. However, water with high concentrations of sulfate can reduce DCAD because the counterion is often Mg or Ca. This may be problematic when aiming for DCAD targets in prefresh diets. For example, assuming water did not provide additional Na or K, if a dry cow consumed 11 kg DM and drank 35 L water per day that contained 500 mg S/L, the S in the water would decrease DCAD by about 90 mEq/kg.

Hardness

Water hardness is usually described as the total cationic effects of Ca and Mg within water, but other cations may exist in water and include zinc (Zn), iron (Fe), strontium, aluminum, and manganese (Mn). Categories of hardness as described by NRC (1980) are soft (0 to 60 mg/L), moderately hard (61 to 120 mg/L), hard (121 to 180 mg/L), and very hard (>180 mg/L). Based on tests of up to 290 mg/L, hardness of water has been observed to have no effect on water intake of lactating cows (Graf and Holdaway, 1952; Blosser and Soni, 1957) but has been observed to be negatively associated with FWI in weaned calves (Senevirathne et al., 2018). However, water hardness may affect water handling systems because it may increase the accumulation of scale and may negatively affect water delivery systems (NRC, 2012). In addition, in-

creasing hardness may reduce cleaning efficiency of milking equipment, and hardness poses a risk factor for bacteriological quality of bulk tank milk (Elmoslemany et al., 2009).

pH

Currently, no guidelines for pH exist for drinking water for dairy cattle; however, the U.S. Environmental Protection Agency (EPA, 2009) recommends that for human consumption, water pH should be between 6.5 and 8.5. No information in the literature was found on the effects of varying the pH of drinking water on water intake, animal health, animal production, or the microbial environment in the rumen. However, pH likely has an influence on the survival of some microorganisms found in water (Szewzyk et al., 2000).

Minerals and Ionic Constituents of Water

Water may contain minerals, which can help meet the mineral requirements of animals, but if concentrations are excessive, these minerals can reduce water intake and have other detrimental effects on health and production. Water can supply absorbable minerals to cows, but generally this does not need to be included in supply calculations because the mineral content of water in most studies that evaluated

mineral nutrition was not measured or considered in supply calculations. Users may consider adjusting dietary mineral supply downward when mineral concentrations in the water being consumed are high. However, including water minerals in total supply usually has a trivial effect on total supply (Castillo et al., 2013).

Speciation refers to the form of any given element in water. Elements may appear as a hydrated ion, as a neutral molecule, as a complex with an additional ion, or as some other molecule. Ground water commonly contains mineral species as hydroxo and carbonate complexes. The reactivity, toxicity, and bioavailability of mineral elements found in water are dependent on the form in which they exist; consequently, simply knowing the concentration of a particular mineral in drinking water yields limited information (NRC,

2005). Table 9-4 lists the major and minor ionic species commonly found in ground water. These species are usually present in water due to contact between water and nearby mineral deposits, while the minor constituents, ammonium, carbonate, and sulfide, may be present because of microbial and algal activity (Tchobanoglous and Schroeder, 1985). Traditionally, water analysis focuses on the total concentration of a mineral in a water sample and usually does not report data related to speciation. Such results can be evaluated for completeness and accuracy by determining if the sum of cations (eq/L) equals the sum of anions (eq/L). This is because, by the principle of electroneutrality, they must be equal in a solution. NRC (2005) notes that the difference of up to 2 percent may be due to uncontrollable error, but a difference

TABLE 9-4 Elements and Major and Minor Ionic Species That Are Common of Ground Waters^a

Major Ionic Species	
Cations	Anions
Calcium (Ca ²⁺)	Bicarbonate (HCO ₃ ⁻)
Magnesium (Mg ²⁺)	Sulfate (SO ₄ ²⁻)
Sodium (Na ⁺)	Chloride (Cl ⁻)
Potassium (K ⁺)	Nitrate (NO ₃ ⁻)
Minor Ionic Species	
Cations	Anions
Aluminum (Al ³⁺)	Bisulfate (HSO ₄)
Ammonium (NH ₄ ⁺)	Bisulfite (HSO ₃ ⁻)
Arsenic (As ³⁻)	Carbonate (CO ₃ ²⁻)
Barium (Ba ²⁺)	Fluoride (F ⁻)
Boron (BO ₃ ³⁻)	Hydroxide (OH ⁻)
Copper (Cu ²⁺)	Phosphate, mono (H ₂ O ₄ ⁻)
Iron, ferrous (Fe ²⁺)	Phosphate, di (HPO ₄ ²⁻)
Iron, ferric (Fe ³⁺)	Phosphate, tri (PO ₄ ³⁻)
Manganese (Mn ²⁺)	Sulfide (S ²⁻)
	Sulfite (SO ₃ ²⁻)

^aAdapted from Tchobanoglous and Schroeder (1985).

of 5 percent or greater suggests error in either sampling or analysis or that one or more ionic species were not reported.

Sulfate

Sulfates in drinking water usually originate from the dissolution of sulfate-bearing minerals located in both soils and rocks. Another source of sulfate contamination in water may be household or industrial wastes and detergents that contain sulfates (Veenhuizen and Shurson, 1992). Laboratories can report either sulfate or sulfate-sulfur and to convert sulfate into sulfate-sulfur multiply by 0.33. High concentrations of sulfate (SO₄²⁻) ions in drinking water may negatively affect both feed and water intake (Loneragan et al., 2001). Weeth and Hunter (1971) observed that when sulfate in drinking water was increased to 3,493 mg sulfate/L (by adding sodium sulfate), water intake by Hereford heifers was reduced by 35 percent. Hereford heifers consuming water with 2,814 mg/L sulfate (from sodium sulfate) reduced feed intake and weight gain (Weeth and Capps, 1972). Although Digesti and Weeth (1976) concluded the safe maximum concentration for sulfate in drinking water is 2,500 mg sulfate/L, the current consensus recommendation is that water sulfate should not exceed 500 mg/L and 1,000 mg/L for calves and adult cows, respectively. NRC (2005) suggests that water for cattle fed high-concentrate diets should contain less than 600 mg sulfate/L while also noting that when consuming a high-forage diet, cattle can safely drink water containing 2,500 mg sulfate/L. Deep well water in some areas may contain 3,000 mg/L or more sulfate (Patterson and

Johnson, 2003). The rumen is a reducing environment; thus, most sulfur (S) originating from salts is reduced to sulfide. This can become so abundant that the combined S from feed and water will together tie up many trace minerals, making them unavailable to the animal. Depending on dietary S concentration, water with 1,000 to 1,500 mg sulfate/L may cause antagonism of copper (Cu) and selenium (Se) (see Chapter 7 for more detail).

Common forms of sulfate in water include Ca, Fe, Mg, Mn, and Na salts. Although the animal's response to increasing sulfate in water would depend on the specific form of sulfate present, little research exists in comparing these forms. In a study using Angus heifers, GROUT et al. (2006) observed that the extent of aversion to water high in sulfate is, in part, dependent on the associated cation. Specifically, they found that increasing the concentrations of sulfate at 1,500, 3,000, and 4,500 mg/L in the form of magnesium sulfate reduced water intake, but reductions were not observed when cattle consumed sodium sulfate.

Iron

Waters containing high concentrations of Fe are often easy to recognize, as the water appears rusty in color, contains sediment, and possesses a metallic taste. Consumption of excessive amounts of Fe can antagonize cobalt (Co), Cu, Mn, Se, and Zn (Olkowski, 2009). Some experimental evidence suggests that oxidative stress may be spurred by high concentrations of Fe in drinking water. Free Fe catalyzes reactions via the Haber-Weiss reaction (Kehrer, 2000). This condition may be brought about when the consumption of Fe exceeds requirements, and as a result, the concentration of reactive oxygen and nitrogen (N) species increases. For example, abomasal infusions of ferrous lactate have been shown to negatively affect milk protein composition and overall stability of milk (Wang et al., 2016). Additional oxidative stress may be of concern in periparturient cows with a compromised immune system (Celi, 2010; Konvicna et al., 2015). Dietary Fe supplementation is rarely needed for adult cattle, and if water contains Fe, dietary supplementation should usually be avoided. The maximum contaminant level of Fe in drinking water for humans is 0.30 mg/L (EPA, 2009), and this concentration is often listed as a caution level for dairy cattle (Genther and Beede, 2013). In a study with sheep, no differences in FWI were observed when the concentration of Fe (from ferric sulfate [Fe₂(SO₄)₃]) was increased from 75 to 145 mg/L (Horvath, 1985). The effect of different Fe concentrations, different valences (ferrous [Fe⁺²] or ferric [Fe⁺³]), and different Fe sources (salts) in drinking water on FWI by lactating dairy cows were tested by Genther and Beede (2013). When water contained added ferrous lactate (Fe(C₃H₅O₃)₂), cows reduced FWI and spent less time drinking with 8 mg/L, compared with 4 or 0 mg/L. Valence of Fe source, namely ferrous sulfate (FeSO₄) or fer-

ric sulfate ($\text{Fe}_2(\text{SO}_4)_3$), did not affect FWI when offered at 0 or 8 mg/L, despite some visual differences in the appearance of the water. When FWI was compared with 0, 8, or 12.5 mg/L from different salts of Fe (ferrous chloride [FeCl_2] or ferric chloride [FeCl_3]), no differences in FWI were observed. When water with 0 or 8 mg/L of ferrous lactate ($\text{Fe}(\text{C}_3\text{H}_5\text{O}_3)_2$), ferrous sulfate (FeSO_4), or ferrous chloride (FeCl_2) was offered, cows drank more water without added Fe, but FWI was not affected by the different ferrous salts. These authors also noted that analytical method had a major effect on assayed Fe concentrations. A direct metal analysis of the raw water sample, without acidification, yielded values that were only 7 to 25 percent of the concentrations obtained when nitric acidification was conducted prior to analysis. Hence, when evaluating data, it is important to know which method was used.

Nitrate

Nitrate (NO_3^-) in drinking water may be a result of industrial pollution or heavy fertilization of fields, or it may be associated with shallow wells (Wang et al., 1999; Wright,

2007). There are currently no documented needs of dietary NO_3^- or nitrite (NO_2^-) by animals (NRC, 2005); however, it

has been used to reduce ruminal methane production. Due to their caustic action, NO_3^- consumed in high concentrations may cause gastroenteritis. In addition, when consumed by cattle, NO_3^- can be used as a source of N for bacteria in the

rumen (Russell, 2002). Most critically, the rumen is also the site of reduction of NO_3^- to NO_2^- . In the case of acute

toxicosis, NO_2^- is absorbed into the bloodstream, which triggers oxidation of the ferrous Fe in hemoglobin to form methemoglobin. This reaction reduces the oxygen-carrying capacity of blood and may cause asphyxiation. Symptoms of NO_2^- poisoning include excessive salivation, abdominal pain, diarrhea and vomiting, and brown-colored mucous membranes (Radostits and Done, 2007). NO_2^- poisoning will result in impaired breathing, gasping, and rapid respiration. Signs may also include muscle tremor, weakness, stumbling gait, cyanosis, and a weak pulse. Abortion in ruminants is believed to follow NO_2^- poisoning (Bruning-Fann and Kaneene, 1993) and has been observed in both dairy and beef herds (Yeruham et al., 1997). In dairy

cows, NO_3^- concentrations up to 180 mg/L in drinking water did not increase the concentration of NO_3^- in milk (Kammerer et al., 1992). In a field study with 54 cows, with half consuming water of 19 mg/L NO_3^- and the other half consuming water of 374 mg/L NO_3^- with the addition of potassium nitrate for 35 months, the first 20 months resulted in no effects on reproductive performance, but in the last 15 months, services per conception increased and first service conception rate decreased in cows drinking the high NO_3^- water, but no differences were observed in blood hemoglobin and methemoglobin (Kahler et al., 1974).

TABLE 9-5 Guidelines for NO_3^- and NO_2^- in Drinking Water^a for Dairy Cattle

Nitrate (NO_3^-) mg/L	Nitrate Nitrogen ($\text{NO}_3\text{-N}$), mg/L	Guidelines
0-44	0-10	Safe for consumption by cattle
45-132	10-20	Generally safe when offered with balanced diets with low nitrate feeds
133-220	20-40	May be harmful if consumed for long periods of time
221-660	40-100	Cattle at risk and possible death
>660	>100	Unsafe—possible death; do not use as water source

^aNitrate nitrogen ($\text{NO}_3\text{-N}$) $\times$ 4.43 = nitrate (NO_3^-).

SOURCE: NRC (1974).

In a survey of 128 Iowa dairy farms, an elevation in the NO_3^- concentration of drinking water was correlated with increasing calving intervals (Ensley, 2000). By increasing the number of NO_3^- metabolizing rumen microbes, ruminants can adapt to diets high in NO_3^- (Allison and Reddy, 1984; Lin et al., 2013).

As in the last publication, $\text{NO}_3\text{-N}$ in water is recommended not to exceed 10 mg/L, which is equivalent to 44 mg/L NO_3^- . Cattle are usually more at risk of NO_3^- poisoning because

of high levels of NO_3^- in feeds, but the concentration in water likely has an additive effect on the animal (ANZECC, 2000). Water testing results, which include NO_3^- and NO_2^- in mg/L, can be converted to N values by dividing these values by 4.43 and 3.29, respectively (NRC, 2005). Table 9-5 lists guidelines for NO_3^- in drinking water of cattle.

Minerals and Potentially Toxic Substances in Water

The tolerable and toxic concentrations of minerals in domestic animals have been reviewed (NRC, 2005). The publication lists guidelines for drinking water for both humans and livestock. The guidelines for humans are listed in Table 9-2. Upper concentration guidelines for cattle are based on those of NRC (2001, 2005), Beede (2012), and Socha et al. (2003) but overall are unique to this publica-

tion. The values included in the table were not developed and reported in attempt to define toxic concentrations or even recommended ranges, but they are intended to be used as a reference when evaluating water samples. The publication notes that although conservative, the EPA enforceable and secondary water quality guidelines can act as safe guidelines for livestock. Enforceable standards are defined as concentrations that cannot be exceeded and set a mark for beyond which action to achieve lower levels must be taken. Secondary standards are concentrations that beyond which cosmetic or aesthetic effects may occur.

Additional points made by the NRC (2005) report that are relevant to feeding dairy cattle include a listing of minerals that fall into five categories. These include minerals that

1. Can be found naturally and at toxic levels in water or may contribute to the overall toxicity of the mineral: most commonly arsenic, barium, Fe, Mn, NaCl, sulfur, and nitrate fall into this category.
2. Can be found naturally and presence is rare but significant risks of toxicities: namely lithium, strontium, and uranium.
3. Usually are found at low levels with toxicity occurring due to the contamination from other sources: aluminum, bismuth, boron, bromine, cadmium, chromium, Co, Cu, lead, mercury, molybdenum, nickel, silicon, tin, and other rare earth elements.
4. Are macroelements unlikely to be found at toxic levels in water but may result in aesthetic secondary effects: Ca, Mg, phosphorus (P), and K.
5. Are trace minerals that may be found in water and may contribute to both a toxic concentration and secondary aesthetic effects: Co, Cu, Fe, Mn, and Se along with water containing a high concentration of NaCl.

Microbiological Considerations of Water

Determination of microorganisms in water is difficult, but drinking water is the largest and most direct source of microbial contaminants and potential pathogens (LeJeune and Gay, 2002). Water may be contaminated by runoff or may be a result of the water distribution and delivery systems. These may enhance bacterial conditions through coatings of biofilms that act as microbial habitats (Van Eenige, 2013). Surfaces of water troughs may also be contaminated by bacteria from cud, fecal matter, dust, feed, or bedding (LeJeune et al., 2001 a,b). Water is commonly evaluated for total coliform bacteria and total fecal coliforms. Total coliforms are a generic group of Gram-negative bacteria. Fecal coliforms are not defined taxonomically and are, as the name suggests, often present in the water because of fecal contamination but may originate from other sources. Fecal coliforms are also known as thermotolerant coliforms (Alonso et al., 1999). Common bacteria found in contaminated water include enteric bacteria *Escherichia coli* and *Salmonella* but may also include other microorganisms such as *Campylobacter jejuni*, *Campylobacter coli*, *Yersinia enterocolitica*, *Yersinia pseudotuberculosis*, *Leptospira*, *Burkholderia pseudomallei*, *Clostridium botulinum*, *Mycobacteria* (pulmonary disease), *Pseudomonas cyanobacteria*, *Cryptosporidium*, and *Giardia* (ANZECC, 2000). As in the last edition of this report, no quality standards are set for water contaminated with microorganisms, as evidence to support them is lacking (Van Eenige, 2013). Water is frequently tested for the presence of

thermotolerant (fecal) coliforms, but this test provides no indication of the presence of microbial pathogens (ANZECC, 2000). In addition, in a study involving feedlot cattle, Sanderson et al. (2005) observed no relationship between water coliform count and fecal prevalence of *E. coli* 0157, but suggested that water coliform count is a measure of *E. coli* 0157 exposure. Despite these limitations, a median threshold for thermotolerant (fecal) coliforms for livestock has been recommended to be 1,000 thermotolerant (fecal) coliforms/L (ANZECC, 2000).

Cows or young stock on pasture may be provided surface water to drink. In these cases, animals may be at risk from toxic cyanobacteria (or blue-green algae). The poisoning of livestock by toxic cyanobacteria was first scientifically reported in the late 1800s when animals consumed water from a freshwater lake at the mouth of the Murray River in South Australia (Francis, 1878). Such mortalities have also been reported in grazing adult dairy cows (Galey et al., 1987; Kerr et al., 1987) and in grazing dairy heifers (Fitzgerald and Poppenga, 1993). It is estimated that 40 of the 2,000 species of cyanobacteria that have been identified are capable of being toxigenic (Briand et al., 2003) and may produce hepatotoxins, neurotoxins, dermatotoxins/irritant toxins, cytotoxins, and toxins that may cause gastrointestinal disturbance (Olkowski, 2009). Colonizing both terrestrial and aquatic biotopes and in both marine and freshwater ecosystems, cyanobacteria are photosynthetic prokaryotes, with growth commonly occurring in late summer to autumn (Briand et al., 2003). Risk factors include shallow waters that are neutral to alkaline (Carvalho et al., 2011) and contain high concentrations of N and P. In many livestock operations, the concentrations of N and P are commonly increased in bodies of water when evaporative losses occur along with manure or fertilizer contaminations (Radostits and Done, 2007). The presence of cyanobacteria is typically determined through microscopic examination. If drinking water is suspected to contain cyanobacteria, an alternative source of drinking water should be made available to cattle until it is treated or determined to be safe (Olkowski, 2009).

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